Q9 – VM placement: problem and model

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Physical machines
  PM    CPU (cores)  RAM (GB)  BW (Gbps)  Power (W)
  PM1            16        32         10        400
  PM2            32        64         20        600
  PM3            24        48         10        500
  PM4             8        16          5        250
  PM5            16        64         10        450

Virtual machines
  VM    CPU (cores)  RAM (GB)  BW (Gbps)
  VM1             4         8          1
  VM2             8        12          2
  VM3             8        16          2
  VM4             2         4          1
  VM5            12        16          3
  Total          34        56          9

Encoding
  One gene per VM = the PM it runs on, e.g. [3, 3, 5, 3, 1] puts VM1, VM2, VM4
  on PM3, VM3 on PM5 and VM5 on PM1. Optimisers search x in [0, 5)^5 and
  decode PM = floor(x) + 1.

Fitness (minimise)
  f = servers_used
    + power_of_active_PMs / (total_power + 1)        tie-breaker, always < 1
    + 100 * (n_violations + sum relative overload)   CPU / RAM / BW limits

Global optimum (exhaustive search of 5^5 = 3125 placements)
  [2, 2, 2, 4, 2]  servers = 2, power = 850 W, fitness = 2.3862
  PM2 (600 W): VM1, VM2, VM3, VM5     CPU 32/32 (100%), RAM 52/64 (81%), BW 8/20 (40%)
  PM4 (250 W): VM4                    CPU 2/8 (25%), RAM 4/16 (25%), BW 1/5 (20%)
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Q9 – comparison of DE, ABC and GA-PSO

30 independent runs per algorithm (seeds 0..29), population 20, budget 2000 fitness evaluations per run.
Utilisation: best placement, mean over active servers. Evals/iters to optimum:
mean over the runs that reached the global optimum (see success rate).

Servers and power

Algorithm	Best servers	Mean servers	Best power (W)	Mean power (W)	Best fitness	Mean fitness	Std fitness
DE	2	2.0	850	871.7	2.3862	2.396	0.0113
ABC	2	2.0	850	856.7	2.3862	2.3892	0.0077
GA-PSO	2	2.0	850	888.3	2.3862	2.4036	0.0163

Server utilisation

Algorithm	CPU util (%)	RAM util (%)	BW util (%)	Mean util (%)
DE	71.9	62.5	30.0	54.8
ABC	62.5	53.1	30.0	48.5
GA-PSO	71.9	62.5	30.0	54.8

Convergence speed and run time

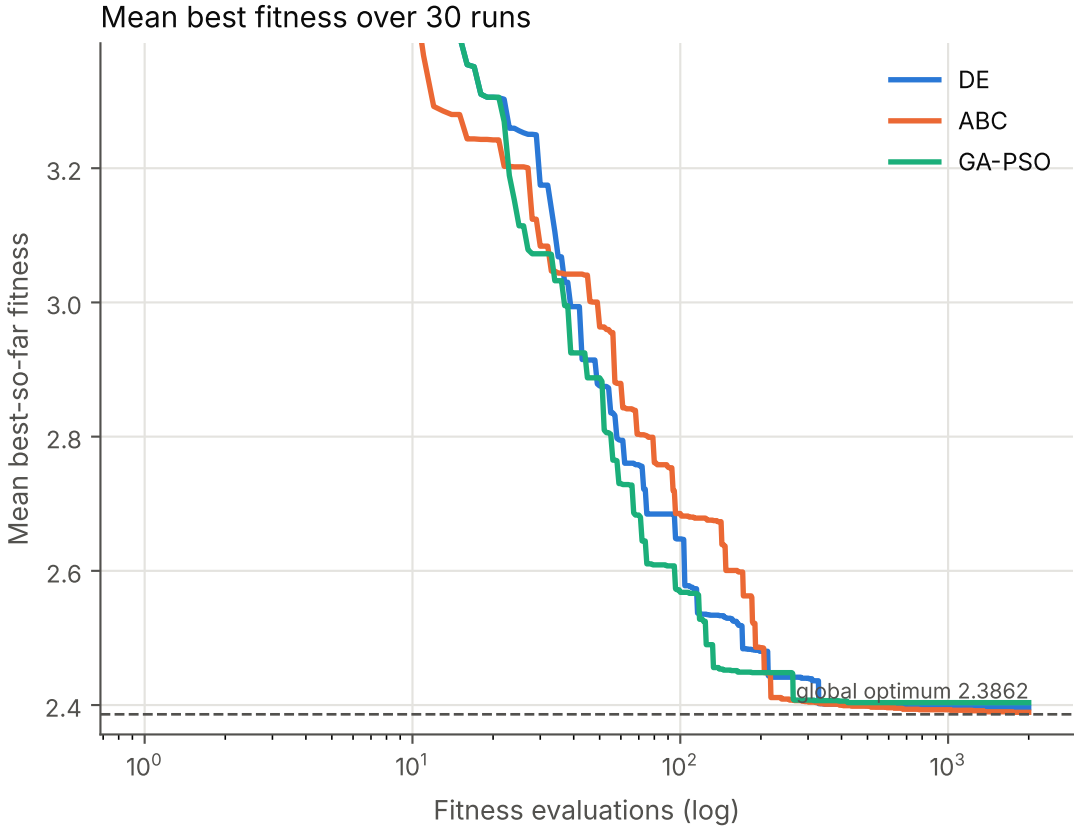
Algorithm	Success rate (%)	Evals to optimum (mean)	Iters to optimum (mean)	Runtime (ms, mean)
DE	56.7	225.1	11.7	166.1
ABC	86.7	525.8	26.8	176.6
GA-PSO	40.0	126.8	6.8	159.3

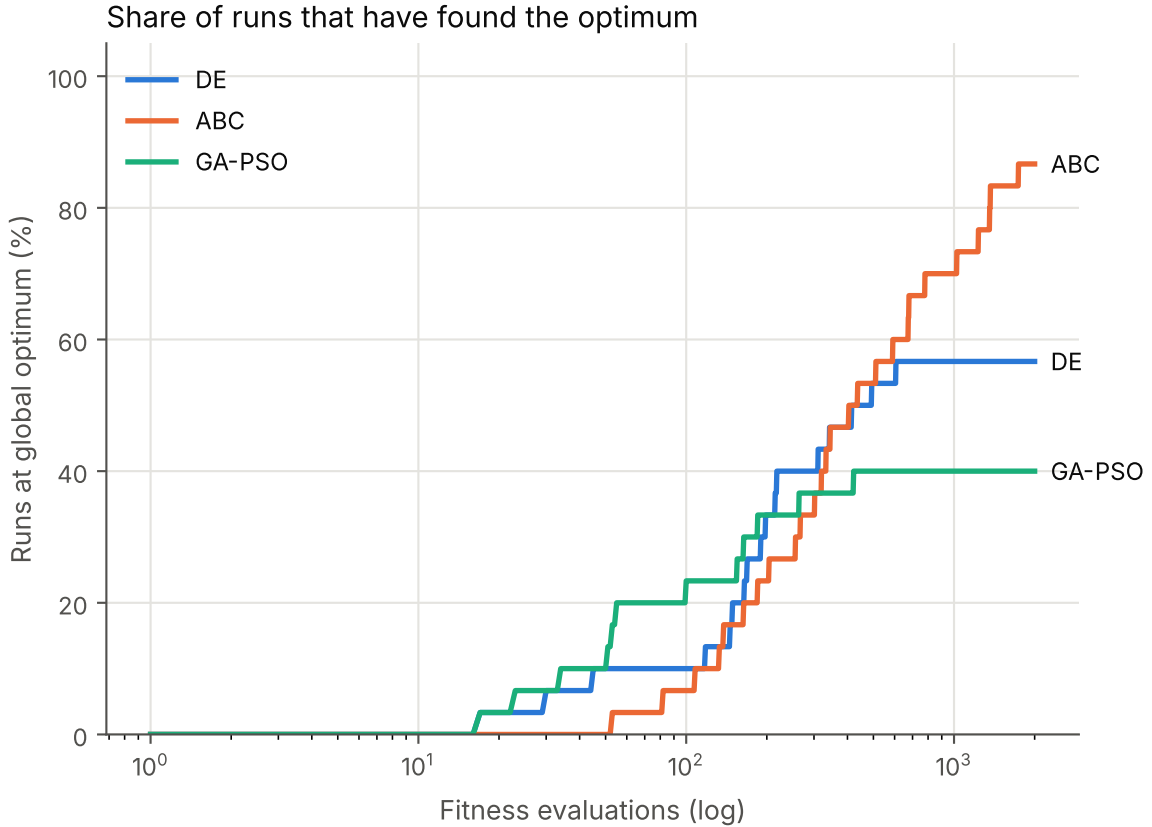
Best placement found by each algorithm

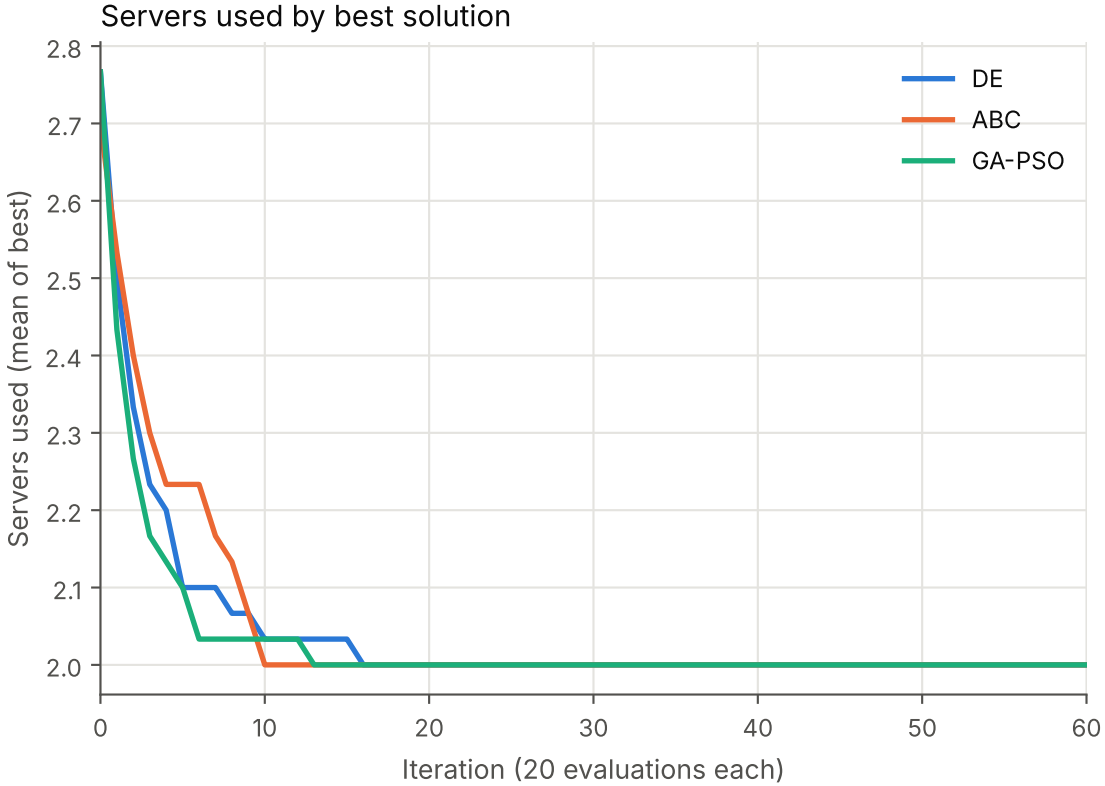
DE: [4, 2, 2, 2, 2] (2 servers, 850 W)
  PM2 (600 W): VM2, VM3, VM4, VM5   CPU 30/32 (94%), RAM 48/64 (75%), BW 8/20 (40%)
  PM4 (250 W): VM1   CPU 4/8 (50%), RAM 8/16 (50%), BW 1/5 (20%)

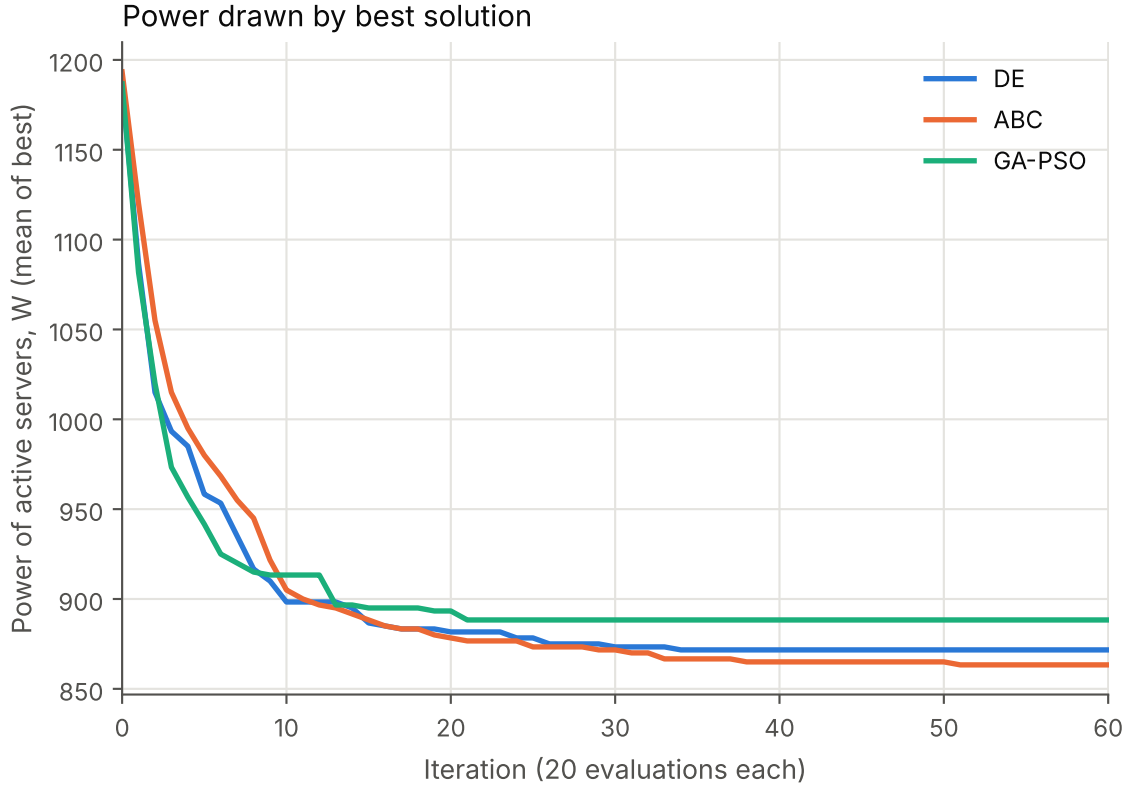
ABC: [2, 2, 2, 4, 2] (2 servers, 850 W)
  PM2 (600 W): VM1, VM2, VM3, VM5   CPU 32/32 (100%), RAM 52/64 (81%), BW 8/20 (40%)
  PM4 (250 W): VM4   CPU 2/8 (25%), RAM 4/16 (25%), BW 1/5 (20%)

GA-PSO: [4, 2, 2, 2, 2] (2 servers, 850 W)
  PM2 (600 W): VM2, VM3, VM4, VM5   CPU 30/32 (94%), RAM 48/64 (75%), BW 8/20 (40%)
  PM4 (250 W): VM1   CPU 4/8 (50%), RAM 8/16 (50%), BW 1/5 (20%)


Mean best fitness over 30 runs
DE
ABC
GA-PSO
3.2
3.0
2.8
2.6
2.4
Mean best-so-far fitness
global optimum 2.3862
10^0
10^1
10^2
10^3
Fitness evaluations (log)


Share of runs that have found the optimum
DE
ABC
GA-PSO
100
80
60
40
20
0
Runs at global optimum (%)
ABC
DE
GA-PSO
10^0
10^1
10^2
10^3
Fitness evaluations (log)


Servers used by best solution
Servers used (mean of best)
2.8
2.7
2.6
2.5
2.4
2.3
2.2
2.1
2.0
0
10
20
30
40
50
60
Iteration (20 evaluations each)
DE
ABC
GA-PSO


Power drawn by best solution
Power of active servers, W (mean of best)
1200
1150
1100
1050
1000
950
900
850
0
10
20
30
40
50
60
Iteration (20 evaluations each)
DE
ABC
GA-PSO